Lecture 6 Practice Questions and Solutions

Question 1 (q1_sol.py)

Write a program that reads a phrase from the user prints sandwiched between two lines of dash characters as shown in the sample run:

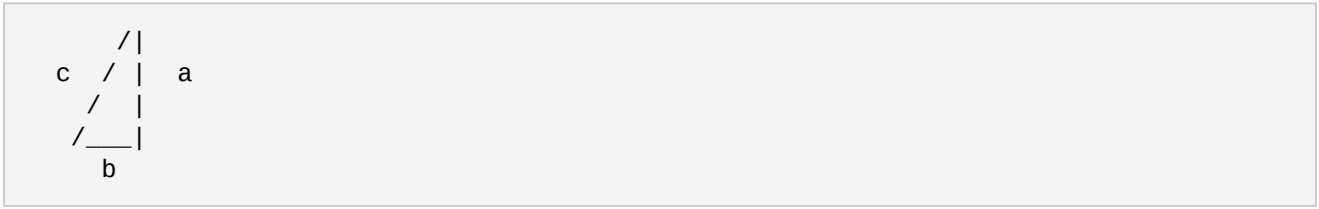
```
Enter a phrase: Hello, world!

-------------
Hello, world!
-------------
```

Sample Solution

```
phrase = input('Enter a phrase: ')
line = '-' * len(phrase)
print(line)
print(phrase)
print(line)
```

Question 2 (q2_sol.py)

Write a program that reads in the lengths of the two legs of a right triangle (a and b in the diagram)and then computes the length of the hypotenuse using the fact that c^2 = a^2 + b^2.



Here's a sample run:

```
What is a? 3.5
What ib b? 6

The hypotenuse c is 6.946221994724902
```

Sample Solution

```
import math

a = float(input('What is a? '))
b = float(input('What is b? '))
c = math.sqrt(a**2 + b**2)
print('The hypotenuse c is', c)
```

Question 3 (q3_sol.py)

Write a program that asks the user how many sides they would like their die to have, and then rolls the die 3 times and prints the results of each roll.

```
How many die sides? 6
Rolling a 6-sided die three times ...
1st roll: 6
2nd roll: 5
3rd roll: 2
```

Sample Solution

```
import random

sides = int(input('How many die sides? '))
print(f'Rolling a {sides}-sided die three times ...')
print('1st roll:', random.randint(1, sides))
print('2nd roll:', random.randint(1, sides))
print('3rd roll:', random.randint(1, sides))
```

Question 4 (q4_sol.py)

Draw a square on the screen using turtle graphics. Make sure the screen does *not* close when the program ends.

Sample Solution

```
import turtle

turtle.forward(200)
turtle.right(90)
turtle.forward(200)
turtle.right(90)
turtle.forward(200)
turtle.right(90)
turtle.forward(200)

turtle.done()
```

Question 5 (q5_sol.py)

Using the tshirt.py module (on the course notes website), draw a t-shirt that says "Please chew quieter". Make it nicely centered.

Sample Solution

```
import tshirt

tshirt.print_line1(' Please')
tshirt.print_line2('  chew')
tshirt.print_line3(' quieter')
```